

Double-Layer Metasurface-based Interface for Dense PIC-to-MCF Polarization-Multiplexed Coupling

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Abstract Efficient and compact edge-coupling interface between SiN photonic integrated circuit (PIC) and multicore fiber (MCF) is demonstrated using a double-layer metasurface device. The PIC-to-MCF coupling losses below 1.4 dB (numerically) and 4.9 dB (experimentally) with polarization-multiplexing functionality are obtained for all eight channels. ©2026 The Author(s)

Introduction

Efficient coupling between multicore fiber (MCF) and photonic integrated circuit (PIC) will be essential for the practical deployment of space-division multiplexed (SDM) transceivers and future high-density co-packaged optics (CPO) using MCF fiber arrays. While grating coupler arrays are conventionally employed to achieve simultaneous coupling from PIC to MCF [1–3], they are inherently sensitive to wavelength and polarization, which would limit their scalability to wavelength-division-multiplexed (WDM) and polarization-division-multiplexed (PDM) systems. To solve this issue, we have recently proposed an efficient interface between edge-coupled PIC and MCF using double-sided dielectric metasurface (MS) [4]. While a preliminary device was demonstrated [5], actual coupling between PIC and MCF has not been achieved due to the insufficient performance.

Here, we experimentally demonstrate, for the first time, a compact MS-based interface between 8-channel edge-coupled SiN PIC and 4-core MCF. Two layers of MS are designed using a newly developed optimization method to simultaneously transform 8-channel output wavefronts from the PIC facet and provide four functionalities—polarization rotation for four

channels, polarization multiplexing, spot-size conversion, and rearrangement of spatial positions—to enable efficient coupling to the MCF. A compact double-sided MS device is fabricated to demonstrate excellent polarization-multiplexed coupling with 3.3~4.9 dB overall PIC-to-MCF loss measured for all 8 channels. By attaching our device directly to the PIC, it offers an alignment-free, detachable MCF connector to a dense PIC without requiring bulky fan-in/fan-out (FIFO) and polarization splitter-rotators.

Device concept and design

Fig. 1 shows the schematic of the MS-based interface demonstrated in this work. MS is a two-dimensional array of subwavelength structures, called meta-atoms, which enable manipulation of the phase and polarization state of transmitted light [6]. By cascading two MS layers, the transmitted light undergoes sequential wavefront transformations, enabling generation of desired output state with sufficient degree of freedom. More specifically, we design spatial distributions of meta-atoms, so that the transverse-electric (TE) mode from each SiN waveguide is selectively coupled to a designated MCF core with a specific polarization state, enabling polarization-multiplexed coupling.

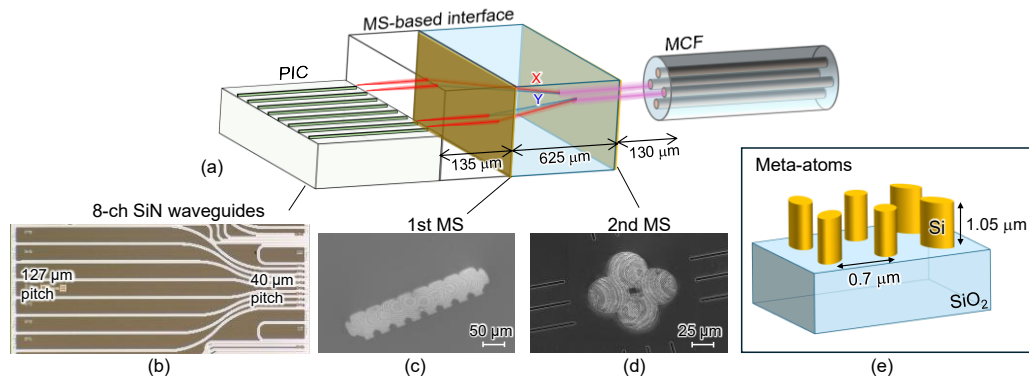


Fig. 1: (a) Schematic of the MS-based interface. 8-channel TE-polarized light from SiN PIC is polarization-multiplexed and coupled to 4-core MCF. (b) Microphotograph of the SiN PIC. (c, d) SEM images of the MS fabricated at both sides of SiO₂ substrate. (e) Structure of meta-atoms.

In this work, we set the operating wavelength to 1550 nm and implement two MS layers by fabricating Si elliptical nanoposts with the height of 1.05 μm and pitch of 0.7 μm on both sides of a 625- μm -thick SiO_2 substrate [Fig. 1(e)]. For convenience of the experiment, we assume an air spacer between the device and the PIC. In practice, however, the MS device could be attached directly to the PIC via another SiO_2 spacer to enable a compact, alignment-free interface to the MCF. The PIC consists of 40- μm -pitch 8-channel SiN waveguides with the thickness of 200 nm and width of 240 nm. We employ a 4-core MCF with a pitch of 39.7 μm and a mode field diameter (MFD) of 10.4 μm . The distance between the PIC and the first MS layer is set to 135 μm , while that between the second MS layer and the MCF is set to 130 μm .

In designing two MS layers, the spatial distributions of meta-atom geometries need to be determined. While the adjoint-based optimization method [7] was employed in our previous work [4,5], it requires iterative optimization with relatively high computational cost. In this work, therefore, we develop a rapid and simple design method by approximating both the input and target optical fields as Gaussian beams, allowing analytical derivation. By imposing the condition to match the target field after propagation through the device, the overall phase distribution is uniquely determined without iterative optimization.

Fig. 2(a) shows the simulated intensity

distributions after transmitting through the designed double-layer MS for 8 different input channels. We confirm that the optical beams from Ch 2, 4, 5, and 7 are converted to Y polarization, polarization-multiplexed to those from Ch 1, 3, 6, and 8, and output to the correct core positions of the MCF with the spot size converted to 9.4 μm . The calculated chip-to-MCF coupling loss is within the range of 1.2~1.4 dB for all channels as shown in Table 1. Due to the expanded spot size, the 1-dB misalignment tolerance for the MCF is as large as 2.25 μm . We can, therefore, attach MCF through a detachable connector without active alignment.

Device fabrication and experiments

The designed MS device was fabricated by combining laser lithography to form the backside alignment markers and EB lithography to pattern MS. We employed an SiO_2 substrate with crystalline Si on one side and amorphous Si on the other side. After depositing Cr layer on the backside as a protective layer, the first MS was fabricated by EB lithography followed by reactive-ion etching. Subsequently, a polymethyl methacrylate (PMMA) layer was coated to protect the first MS, and the second MS was fabricated after removing the backside Cr layer. The scanning-electron microscope (SEM) images of the fabricated MS layers are shown in Fig. 1 (c) and (d).

The SiN PIC with 8-channel 40- μm -pitch waveguide array was fabricated by multi-project wafer (MPW) service provided by Ligentec. To

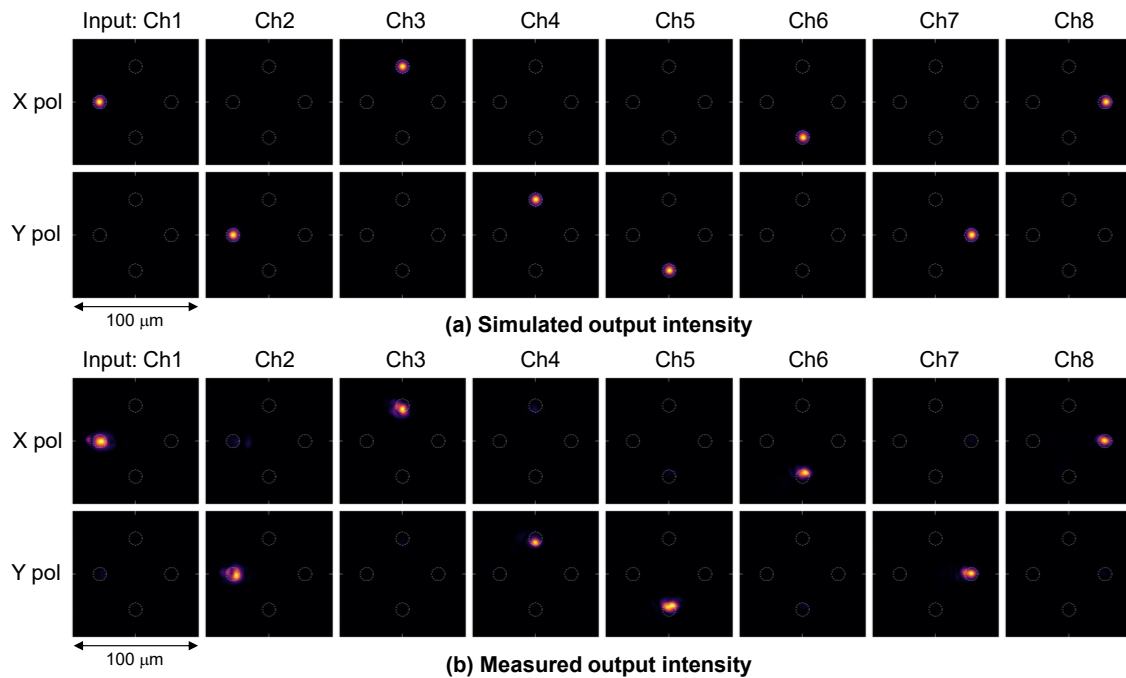
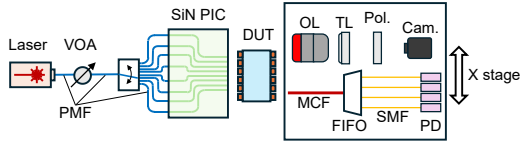


Fig. 2: (a) Simulated and (b) measured intensity distributions of X and Y polarized components after transmitting through the double-layer MS device when TE-mode light from 8 different channels of SiN waveguides is input. Four circles with 10.4- μm diameter (MFD) are drawn by white broken lines at the core positions of MCF for reference.

Table 1: Simulated and measured PIC-to-MCF coupling loss for all input channels.

Input channel	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8
Simulated loss (dB)	1.3	1.2	1.3	1.2	1.2	1.3	1.3	1.4
Measured loss (dB)	4.0	3.3	3.4	4.1	4.0	4.2	4.2	4.9

**Fig. 3:** Experimental setup. VOA: variable optical attenuator, DUT: device under test, SMF: single-mode fiber, PD: photodetector, OL: objective lens, TL: tube lens, Pol.: polarizer, Cam. NIR camera.

ease simultaneous characterization of all channels, pitch-converting waveguides were fabricated as shown in Fig. 1(b), where the opposite side of the chip was coupled to a 127- μm -pitch polarization-maintaining fiber (PMF) array.

Fig. 3 shows the experimental setup to characterize the fabricated MS device. Continuous-wave light at 1550-nm wavelength from a laser source was launched via a PMF to excite the TE mode of each selected channel on the SiN PIC. At the output, the intensity distribution was observed using a near-infrared (NIR) camera after selecting either X- or Y-polarized component by a polarizer. Then, the setup was switched by a translational stage to couple output light from the device to a 4-core MCF to characterize the PIC-to-MCF coupling loss.

Fig. 2(b) shows the measured intensity distributions of X and Y polarization components for all 8 channels. They agree well with the simulated results, demonstrating that both the core position and polarization state are selectively controlled depending on the input waveguide channel. Finally, Table 1 summarizes the measured PIC-to-MCF losses for all channels. They are within a 1-dB range of 3.3~4.2 dB, except for Ch8, which exhibits a slightly higher loss of 4.9 dB possibly due to the larger alignment error. Degradation from the simulated values is attributed mainly to additional scattering caused by residual roughness of the meta-atoms and reflection at the MS interfaces. Further improvement is expected through optimization of the fabrication process and appropriate anti-reflection coating of MS [8].

Conclusions

We have experimentally demonstrated a MS-based interface for direct edge-coupling from PIC to MCF. The fabricated device simultaneously achieved array rearrangement, spot size conversion, polarization rotation, and multiplexing of TE modes output from 8-channel SiN waveguides to 4-core MCF. The overall PIC-to-MCF coupling loss of 3.3~4.9 dB is obtained experimentally for

all channels, with further improvement expected. The demonstrated device could be attached directly to the PIC via another SiO₂ spacer, enabling compact, alignment-free, detachable connection between MCF and PIC for the next-generation SDM transceivers and ultra-dense CPO.

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